KU LEUVEN

Natural Language Processing: Introduction

H02B1A



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Outline

Introduction

State of NLP

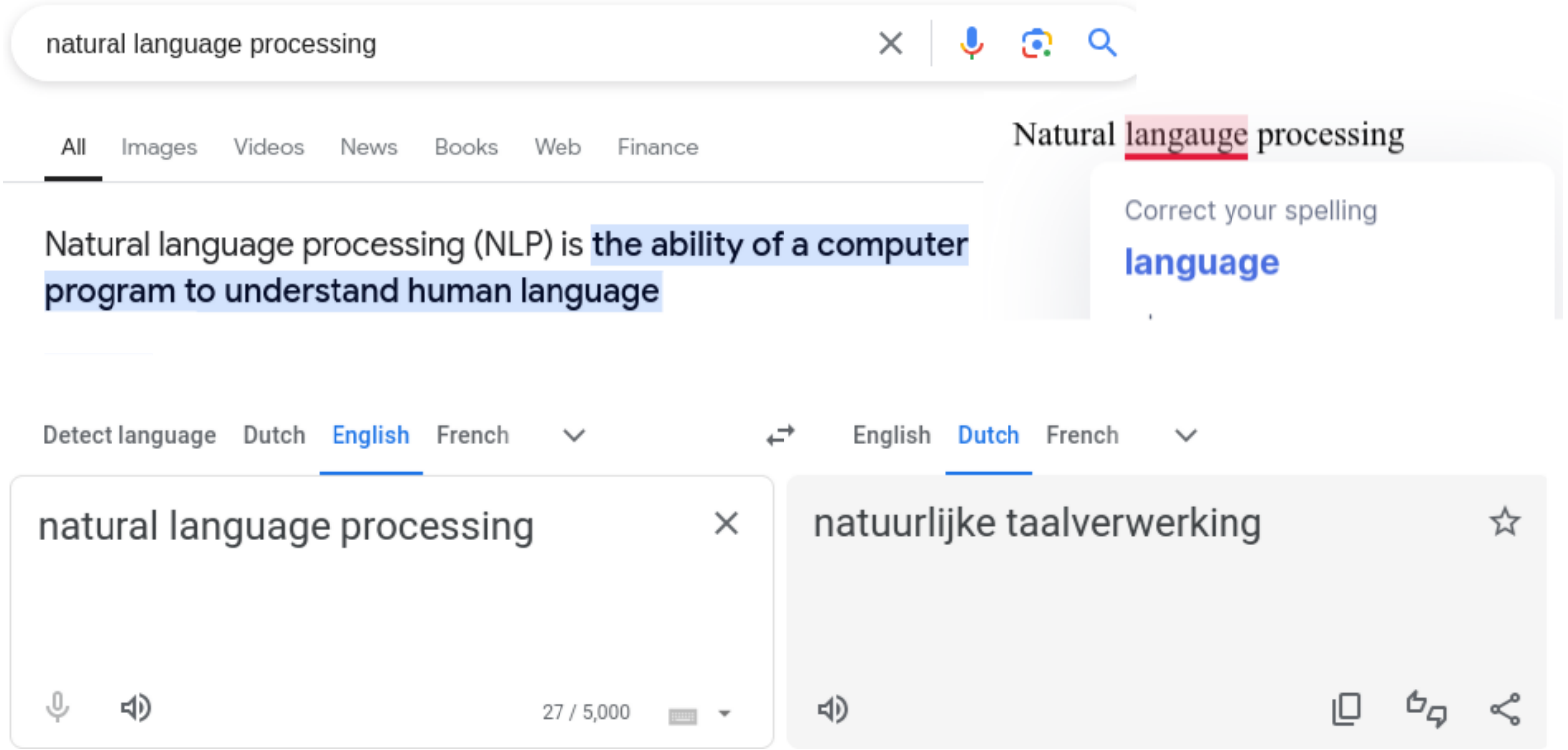
General challenges

Representing words and sentences

Course overview

Practical information

Natural language processing



Natural language processing: definition

Natural language processing is a field of science and engineering focused on the development and study of automatic systems that understand and generate natural (that is, human,) languages.

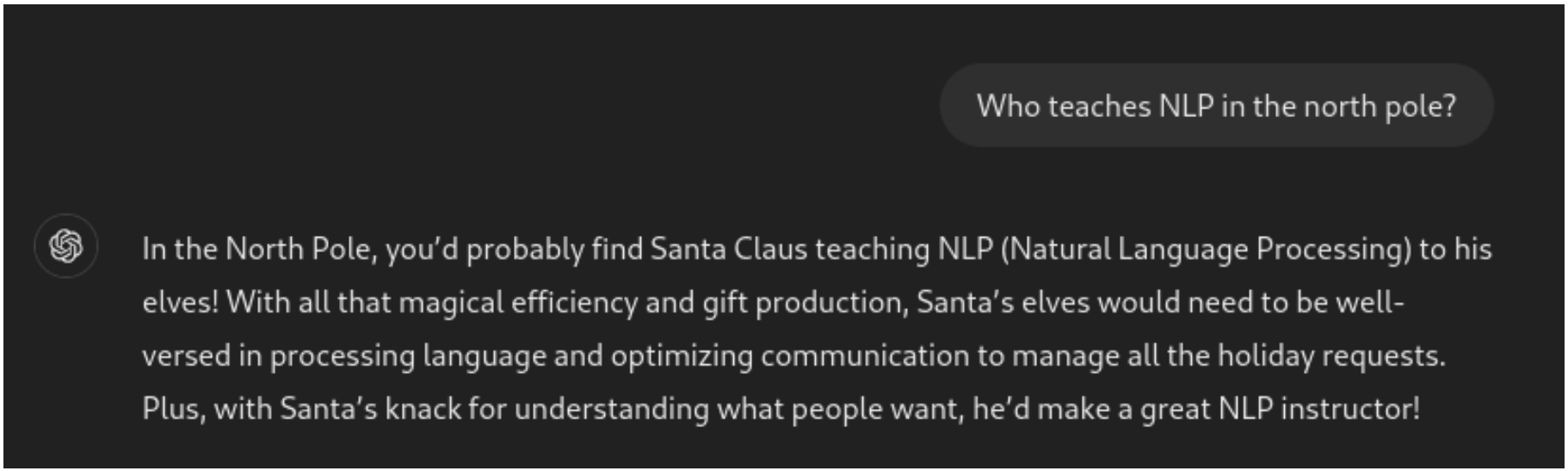
StanfordNLP (2024)

Natural language processing: all-in-one



Retrieved 19/08/2024

Natural language processing: all-in-one



Retrieved 19/08/2024

This course

Foundations

- Building blocks of state-of-the-art NLP systems
- What challenges have been overcome?
- What challenges are still open?

Tasks and applications

- Tasks that cover all levels of linguistic processing (morphology, syntax, semantics)
- Example applications (e.g. machine translation and conversational agents)

This course

Learning goals

At the end of the course, you should be able to

- describe fundamental concepts and algorithms seen in class
- implement and apply these algorithms
- describe and use state-of-the-art NLP models

This lecture

- Current state of NLP
- General challenges
- Representing words and sentences
- Practical aspects of the course

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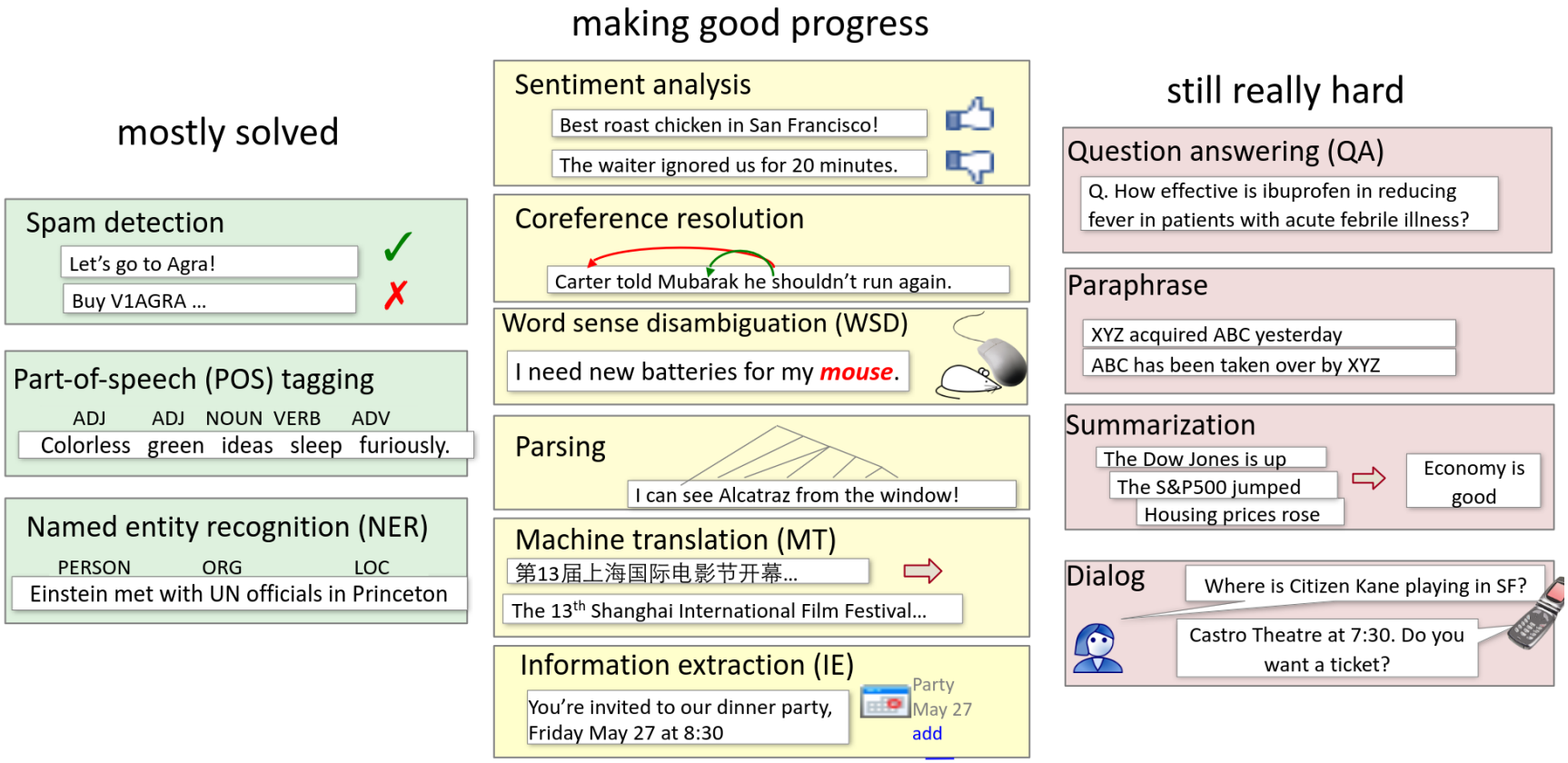
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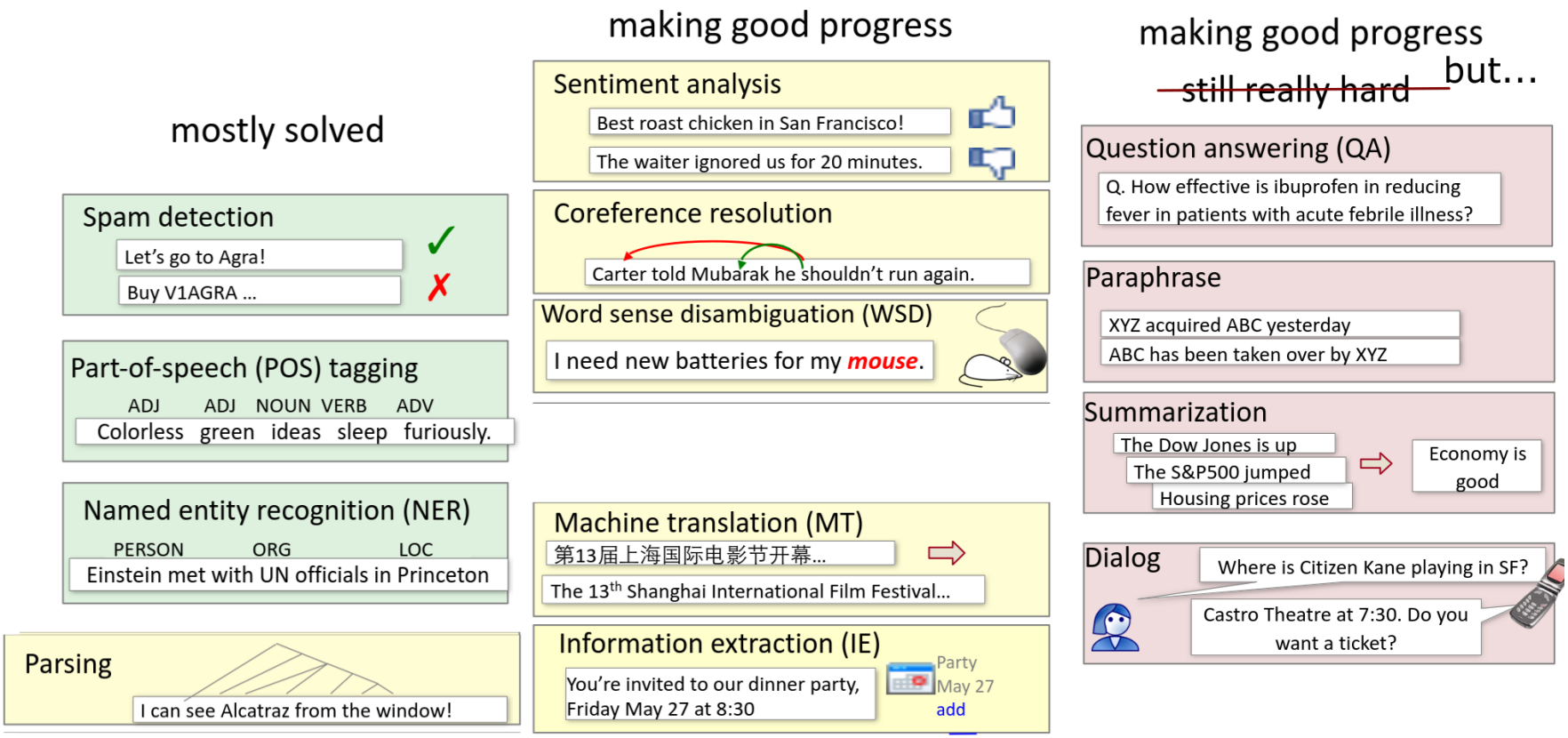
Practical information

State of NLP



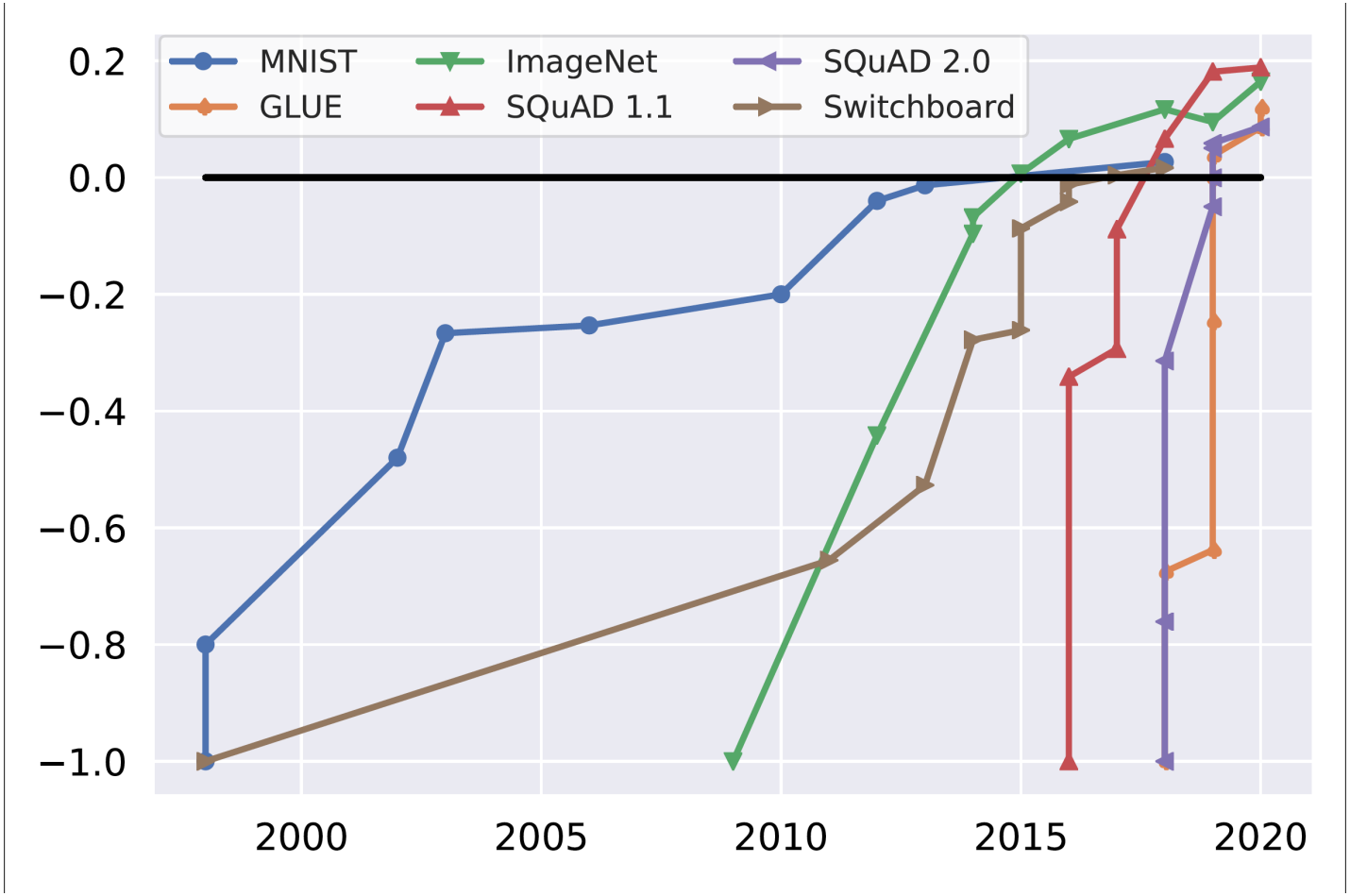
NLP progress in 2012 by Dan Jurafsky

State of NLP



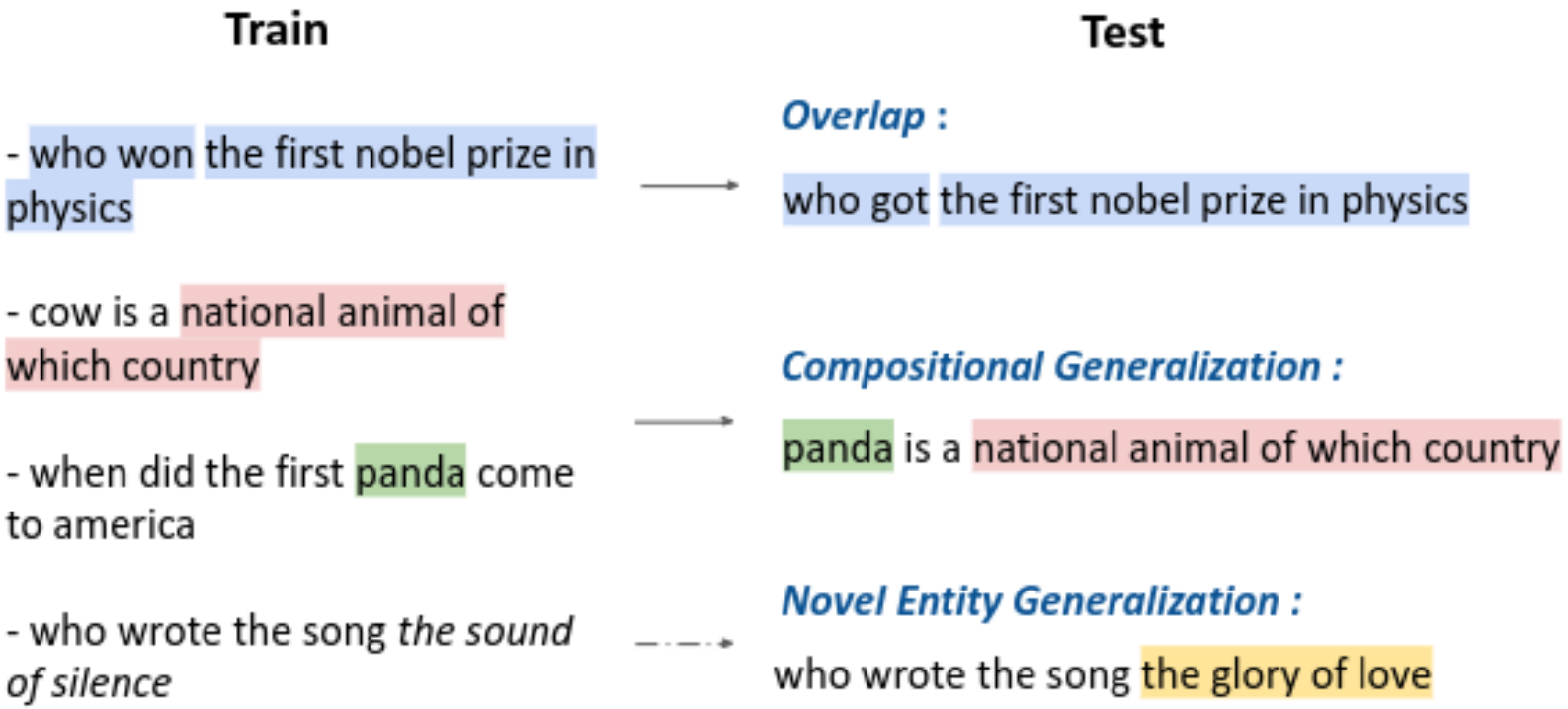
NLP progress in 2025, adapted from Dan Jurafsky

State of NLP



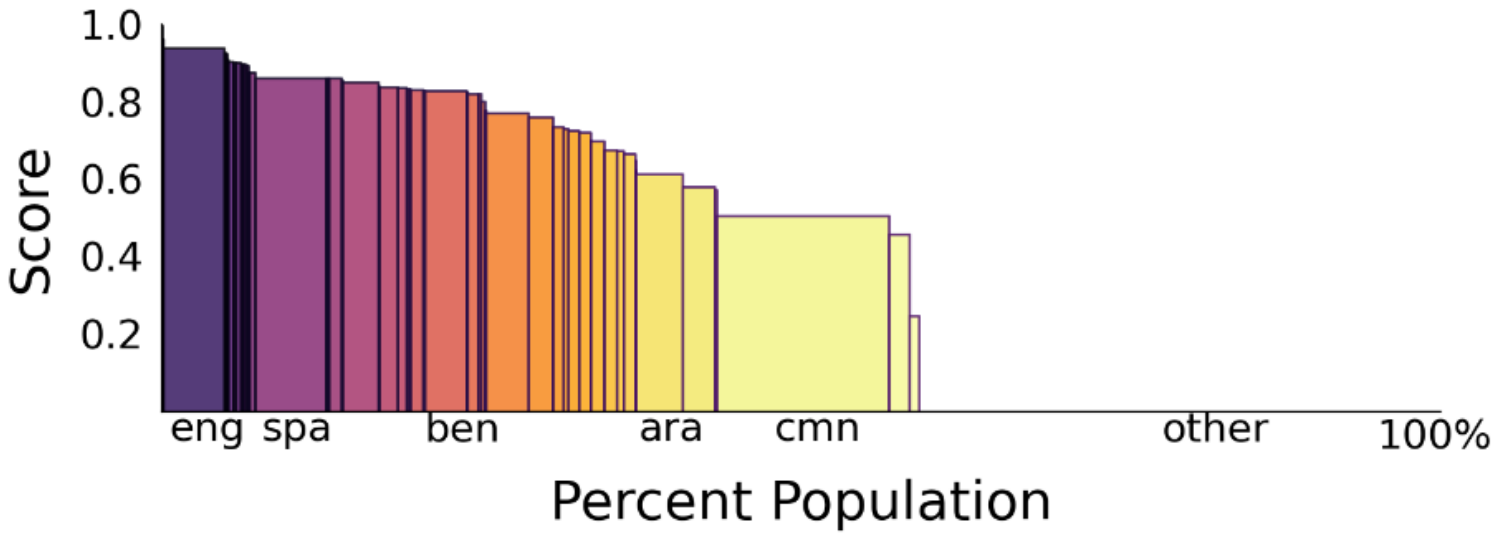
Benchmark progress in NLP and CV. -1=initial performance, 0=human performance. Kiela et al. (2021).

State of NLP



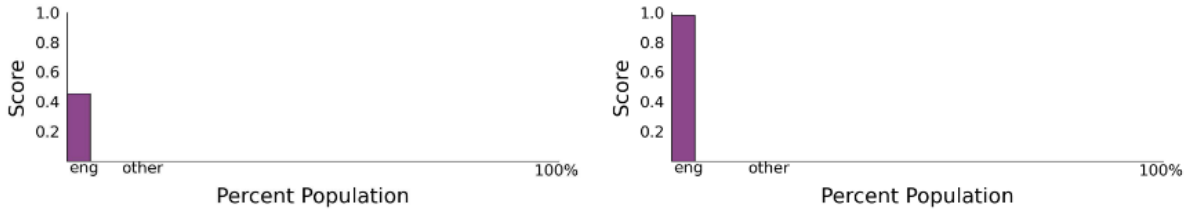
Challenges of open-domain QA (Liu et al., 2022).

State of NLP



Performance disparity of NER systems considering languages and their number of speakers (Song et al., 2023).

State of NLP

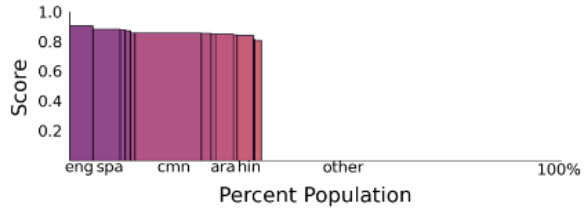


(a) KG Link Tail Prediction

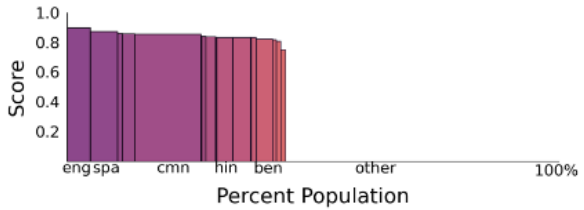
(b) Text Classification



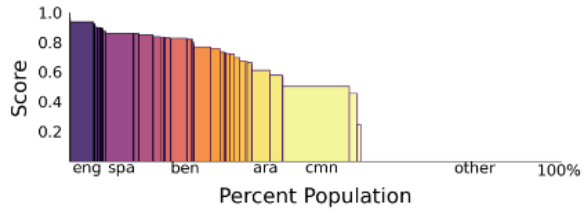
(c) Machine Translation



(d) Text Pair Classification



(e) Extractive QA



(f) Named Entity Recognition

Performance disparity of systems considering languages and their number of speakers for 6 tasks. (Song et al., 2023).

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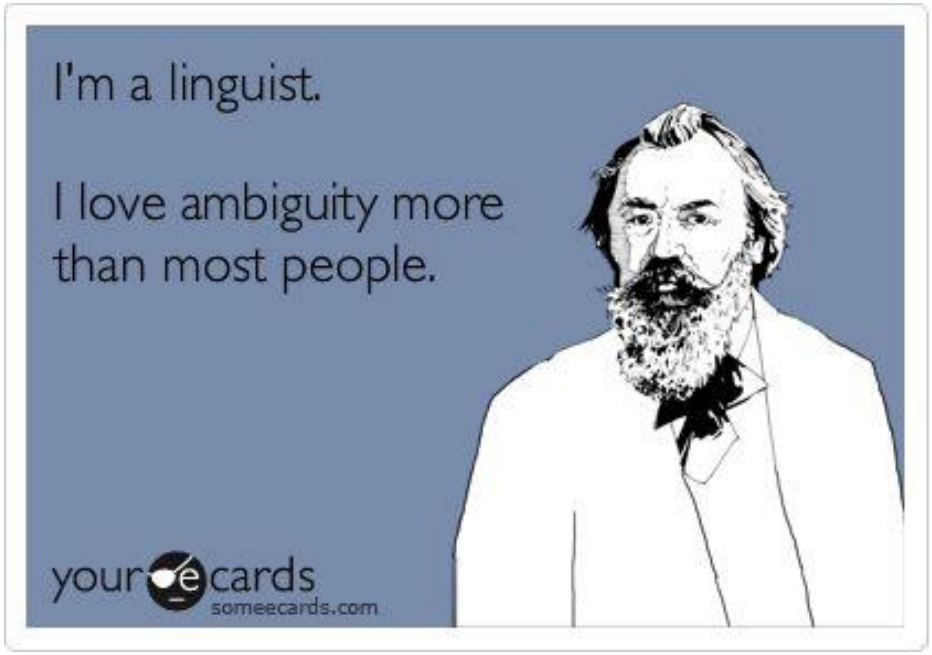
Course overview

Practical information

What makes NLP challenging?

- Ambiguity
- Non-standard language
- Neologisms
- Need for external knowledge
- Long tail of rare words/pheonomena

Ambiguity



- I love ambiguity more than most people love ambiguity.
- I love ambiguity more than I love most people.

Ambiguity

I made her duck

- I cooked duck for her
- I cooked a duck which belonged to her
- I created the duck which belongs to her
- I caused her to lower her head
- I turned her into a duck

Need context to disambiguate.

Ambiguity

Types of ambiguity

- Lexical: *I saw a bat*

  saw = past tense of *see* or present tense of *saw*
  bat = flying mammal or wooden club

- Structural: *I saw a man with a telescope*

  I have a telescope or *the man* does

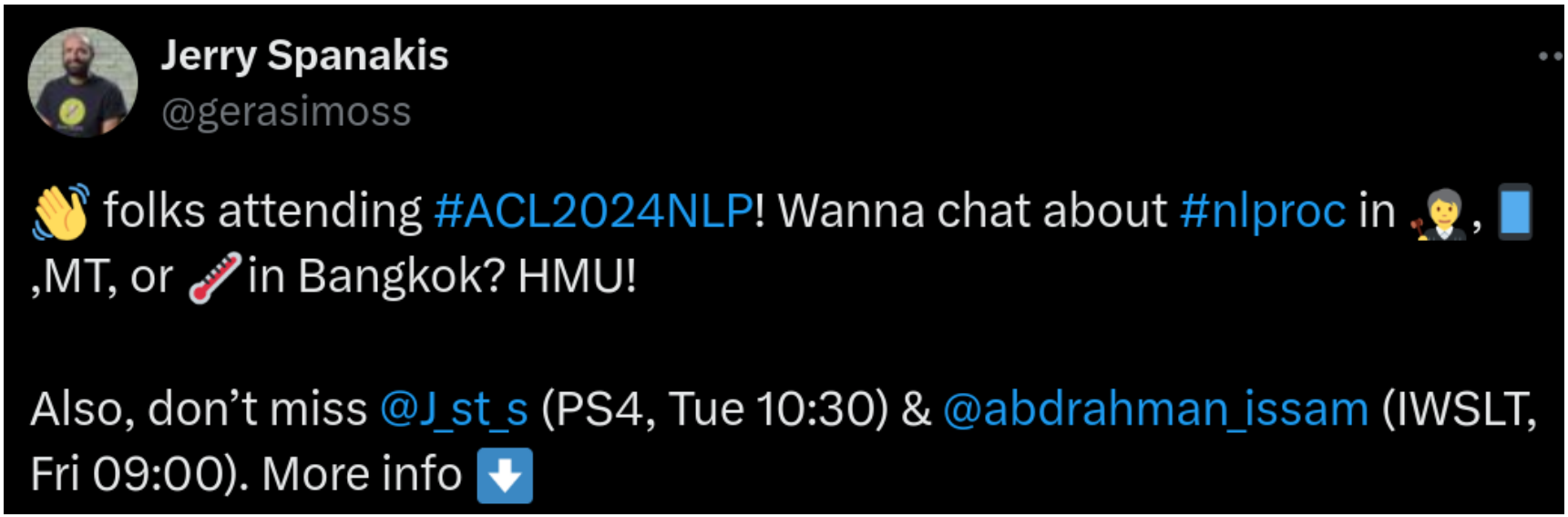
- Semantic: *John and Mary are married*

  to each other or to other people

- Anaphoric: *Margaret invited Susan, and she gave her a sandwich.*

  she = Margaret or Susan

Non-standard language



Example tweet including emoji, hashtags, slang (*wanna*), abbreviations (*MT*, *HMU*)

Neologisms

Recent neologisms

- Deepfake
- Prompt engineer
- Situationship
- Enshittification

Neologisms



NYT first said bluesky bot 🔗

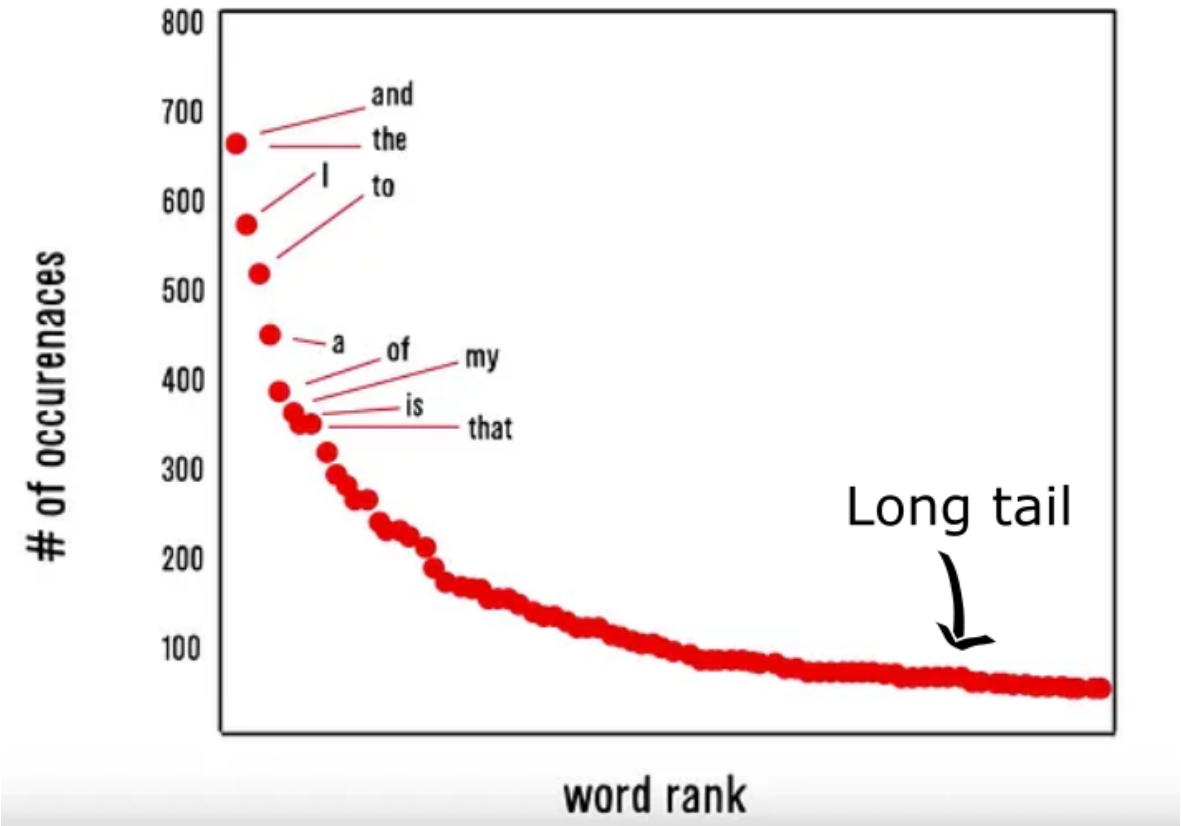
Need for external knowledge

> *Jack needed some money, so he went and shook his piggy bank. He was disappointed when it made no sound.*
>
> Why was he disappointed?
>
> Minsky (2000)

Types of external knowledge

- Commonsense knowledge of the physical and social world
- Domain knowledge
- Knowledge of the broader context

Law of Zipf



Frequency of words per rank in Romeo and Juliet 🔗

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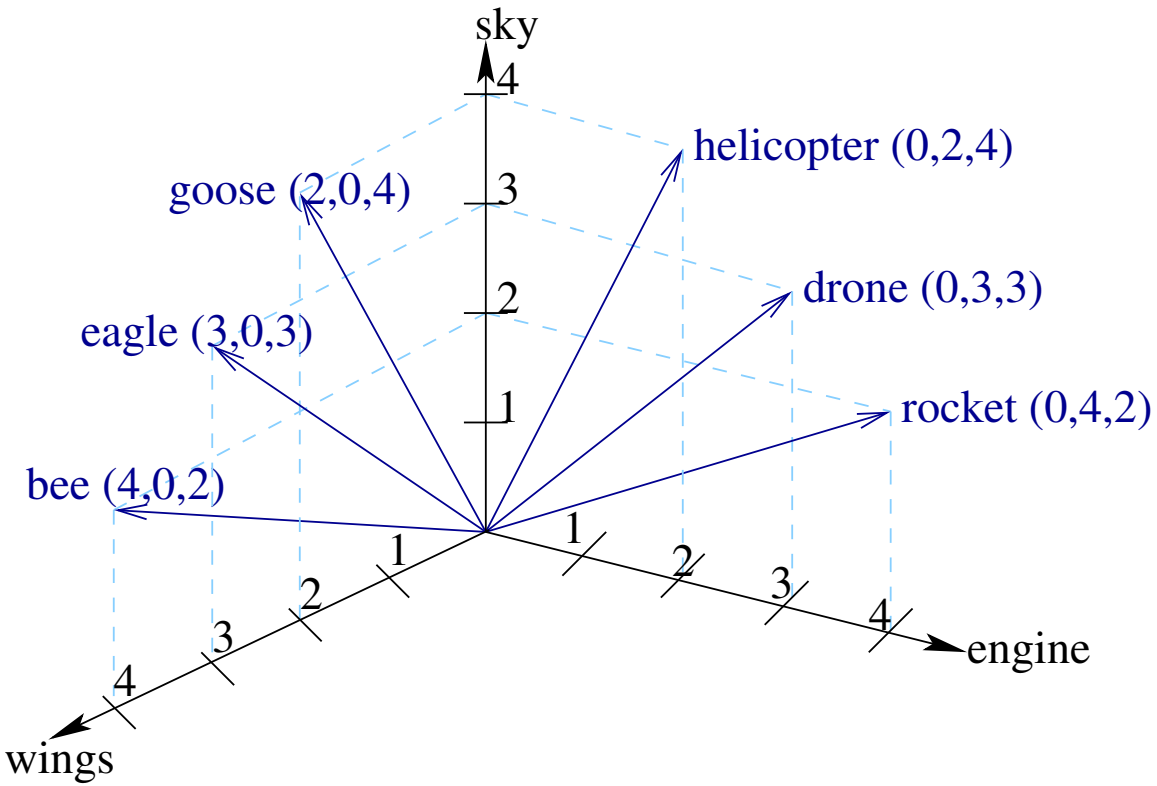
Representing words

Vocabulary: {sky, wings, engine, helicopter, drone, rocket, goose, eagle, bee}

The goose is in the sky.
The eagle in the sky has wings.
The bee in the sky has wings.
The goose with wings in the sky.
The eagle with wings in the sky.
The bee with wings in the sky.
The goose and the eagle with wings are in the sky.
The goose flies in the sky.
The wings of the bee are good wings.
The helicopter, the drone and the rocket are in the sky.
The helicopter in the sky is a useful engine.
The drone in the sky is an engine that takes pictures.
A rocket is an engine.
The helicopter is an engine.
A drone is an engine that takes pictures.
A rocket is a useful engine among sky engines.
A helicopter is flying in the sky.
The sky is full of helicopters and drones.
The engine here is a drone, the one there is a rocket.

Toy corpus

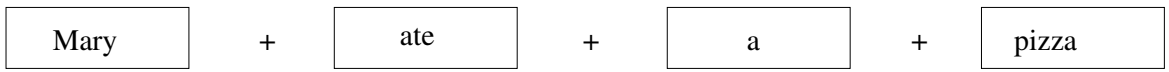
Representing words



Vector space of word co-occurrences with 3 dimensions

Representing phrases/sentences

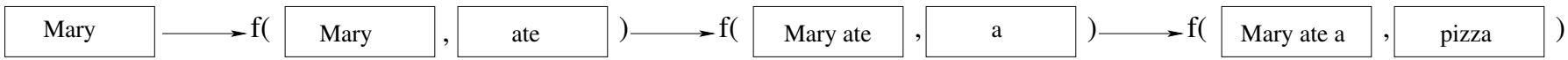
Bag of words



Mary ate a pizza

Representing phrases/sentences

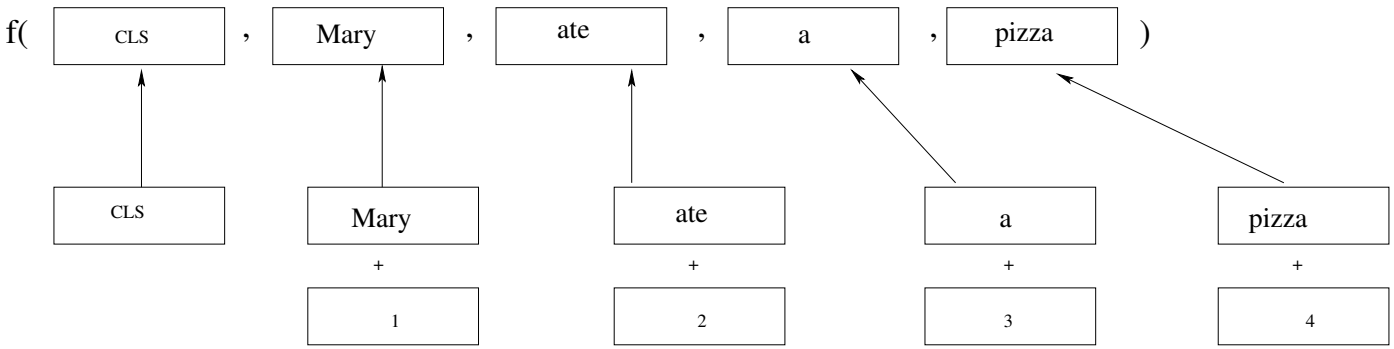
Recurrent Neural Network



Mary ate a pizza

Representing phrases/sentences

Transformer



Mary ate a pizza

Representing phrases/sentences

The reality is not so simple

- Word vectors are learned/updated by the neural network
- $f()$ is complex for RNNs (see lecture 4)
- $f()$ is even more complex for transformers (see lecture 4)
- Neural networks are deep and consist of multiple layers.
- The input units of transformers are not words but subwords (see lecture 2)

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Fundamentals

1. Introduction
2. Segmentation and tokenization
3. Language modelling
4. Neural language modelling

Foundational application

5. Neural Machine Translation

Levels of linguistic processing

6. POS tagging
7. Morphological analysis
8. Syntactic parsing
9. Semantics
10. Discourse

Applications

11. Question Answering
12. *(Not a lecture: Q&A session)*
13. Conversational agents

2. Tokenization

Sample Data:

"This is tokenizing."

Character Level

[T] [h] [i] [s] [i] [s] [t] [o] [k] [e] [n] [i] [z] [i] [n] [g] [.]

Word Level

[This] [is] [tokenizing] [.]

Subword Level

[This] [is] [token] [izing] [.]



- What units to even use?
- How to split text into these units?

3. Language modelling

Before

P(I saw a cat on a mat) =

P(I)
· P(saw | I)
· P(a | I saw)
· P(cat | I saw a)
· P(on | I saw a cat)
· P(a | I saw a cat on)
· P(mat | I saw a cat on a)

After (3-gram)

P(I saw a cat on a mat) =



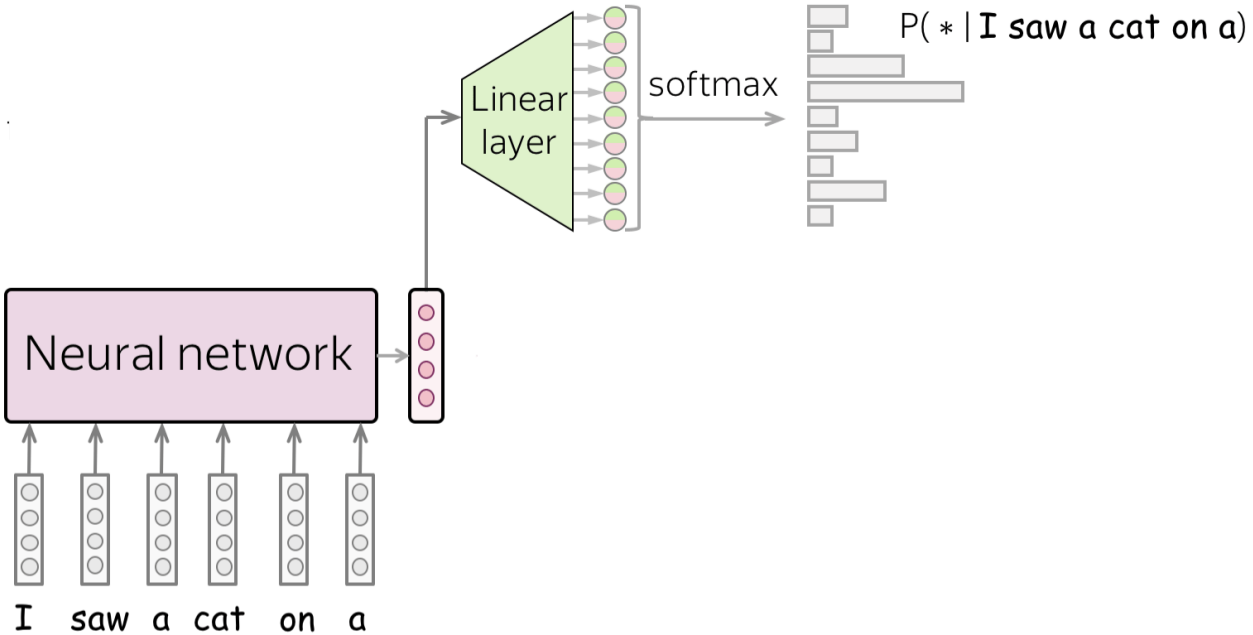
P(I) → P(I)
· P(saw | I) → · P(saw | I)
· P(a | I saw) → · P(a | I saw)
· P(cat | ~~I~~ saw a) → · P(cat | saw a)
· P(on | ~~I saw~~ a cat) → · P(on | a cat)
· P(a | ~~I saw a~~ cat on) → · P(a | cat on)
· P(mat | ~~I saw a cat~~ on a) → · P(mat | on a)

ignore use



Probabilistic foundations of language modelling

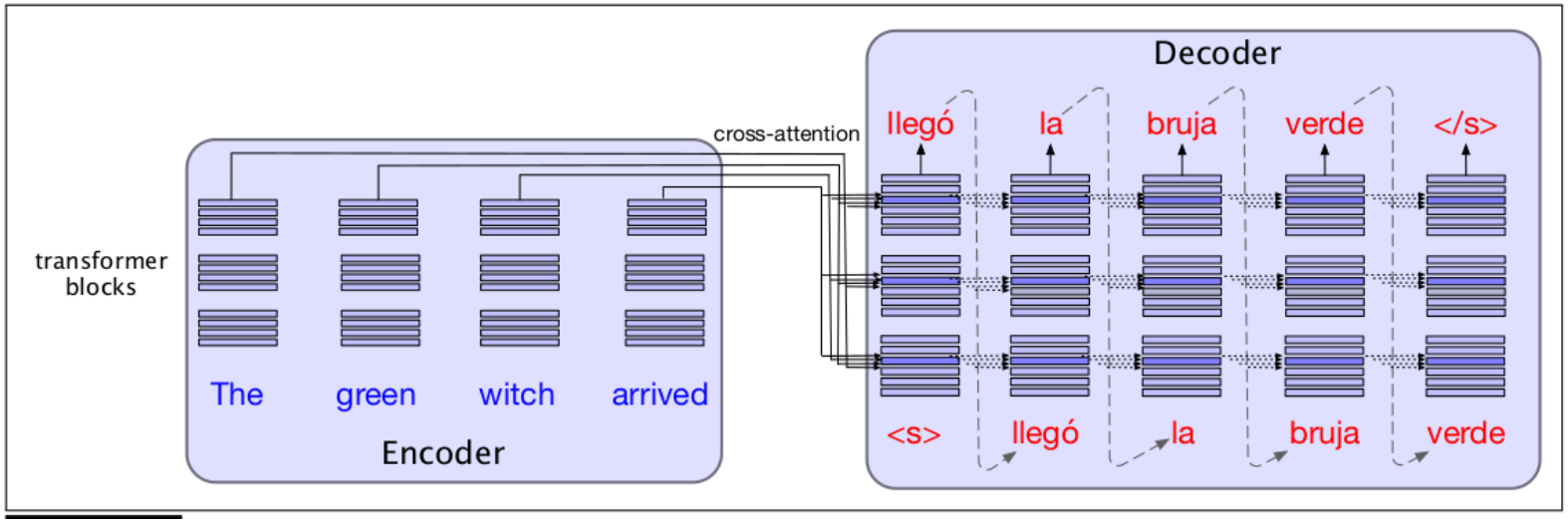
4. Neural Language modelling



- Language modelling with RNNs
- Language modelling with transformers

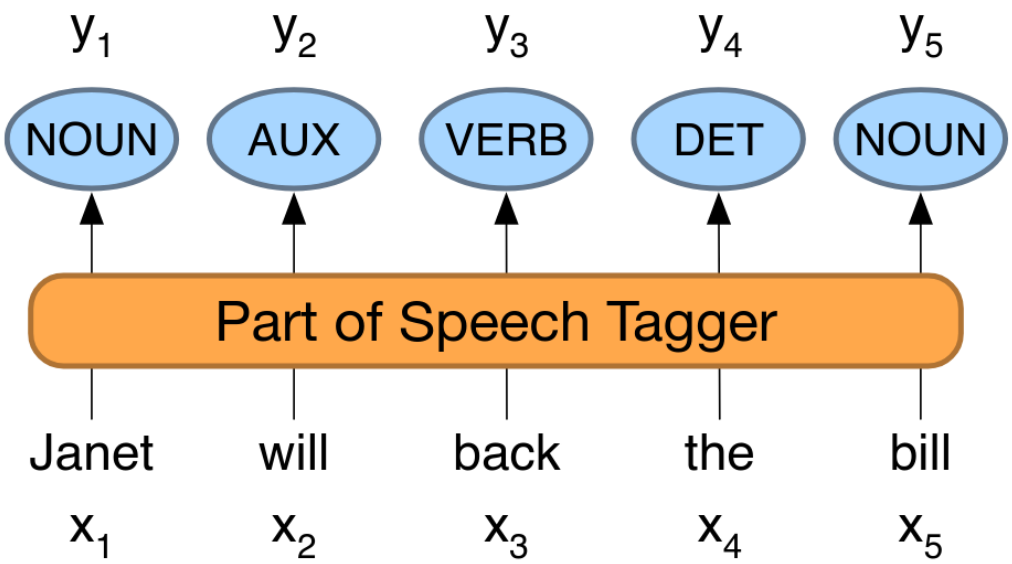
Lecture by Tim Van de Cruys

5. Neural Machine Translation



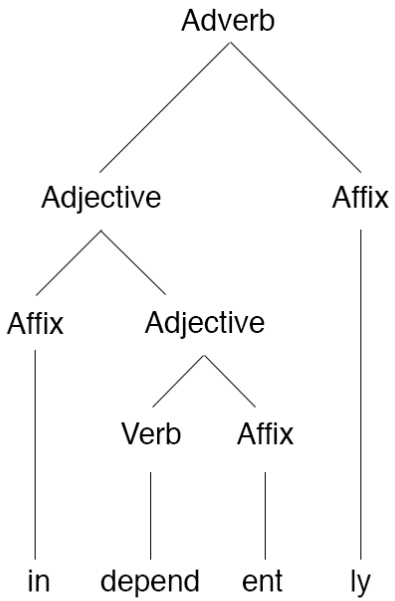
The encoder-decoder transformer architecture for machine translation (Jurafsky and Martin, 2024).

6. POS tagging



POS tagging: mapping from input words $x_1, x_2, ..., x_n$ to output POS tags $y_1, y_2, ..., y_n$ (Jurafsky and Martin, 2024).

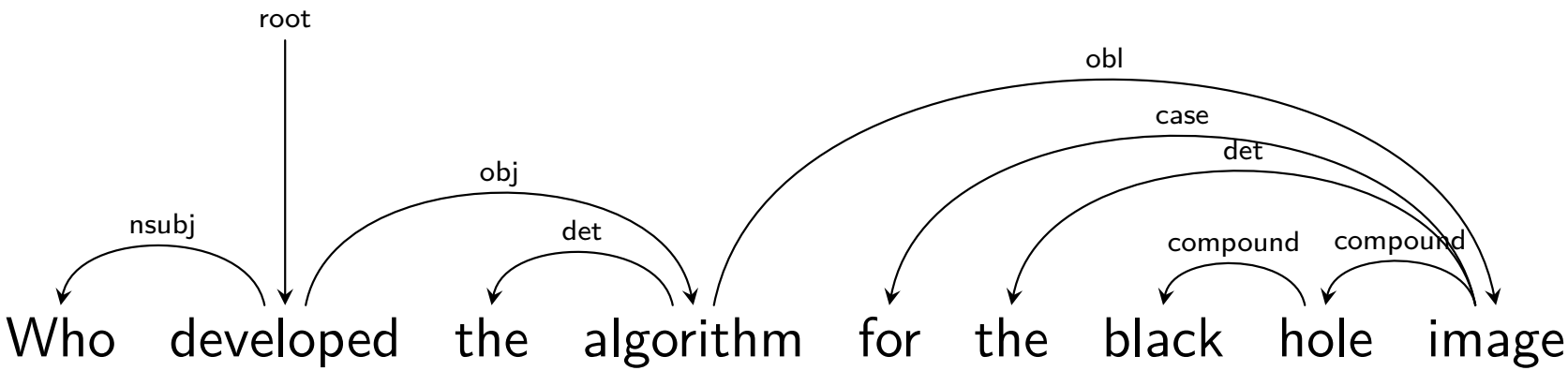
7. Morphological analysis



Morphological analysis of the word *independently* 🔗

Analysing the internal structure of words

8. Syntactic parsing



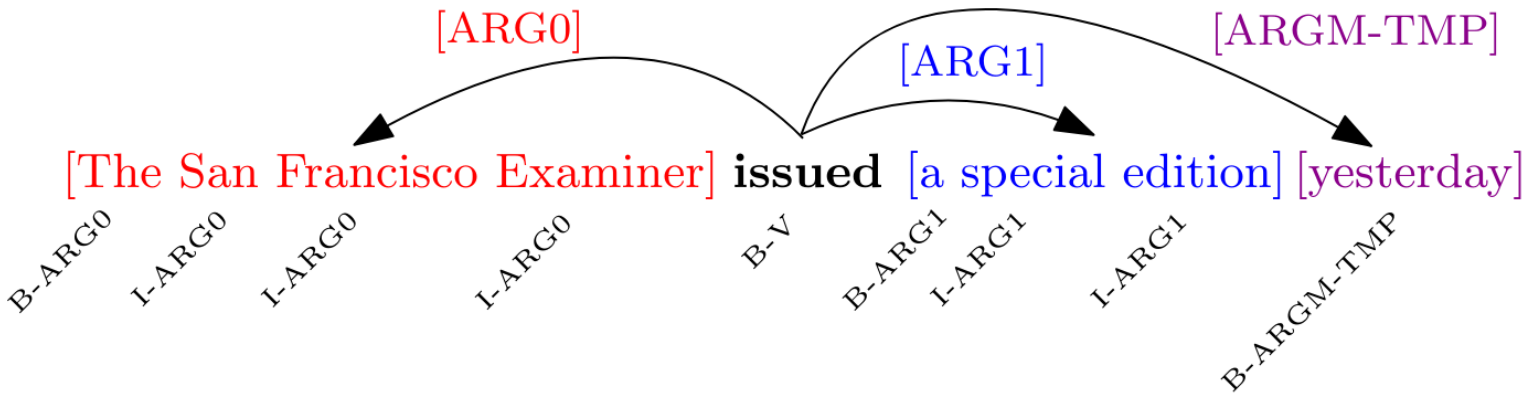
Analysing the structure of sentences

9. Semantics



Word sense disambiguation 🔗

9. Semantics



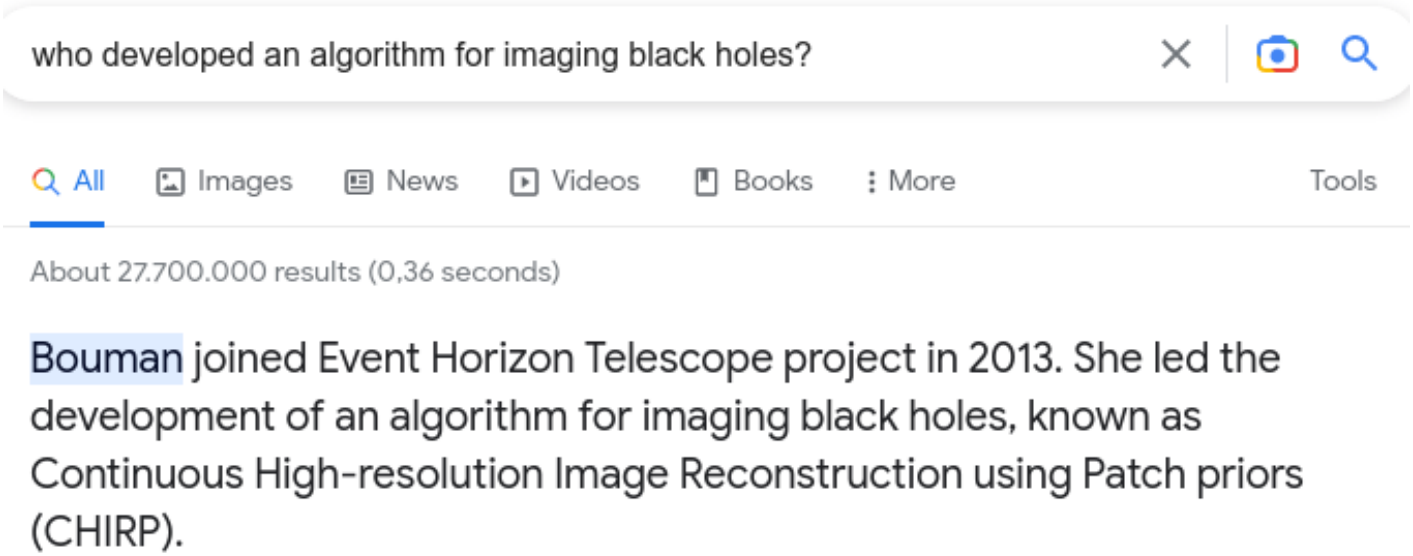
Semantic role labelling: who did what to whom, when (Jindal et al., 2020)

10. Discourse



Coreference resolution 🔗

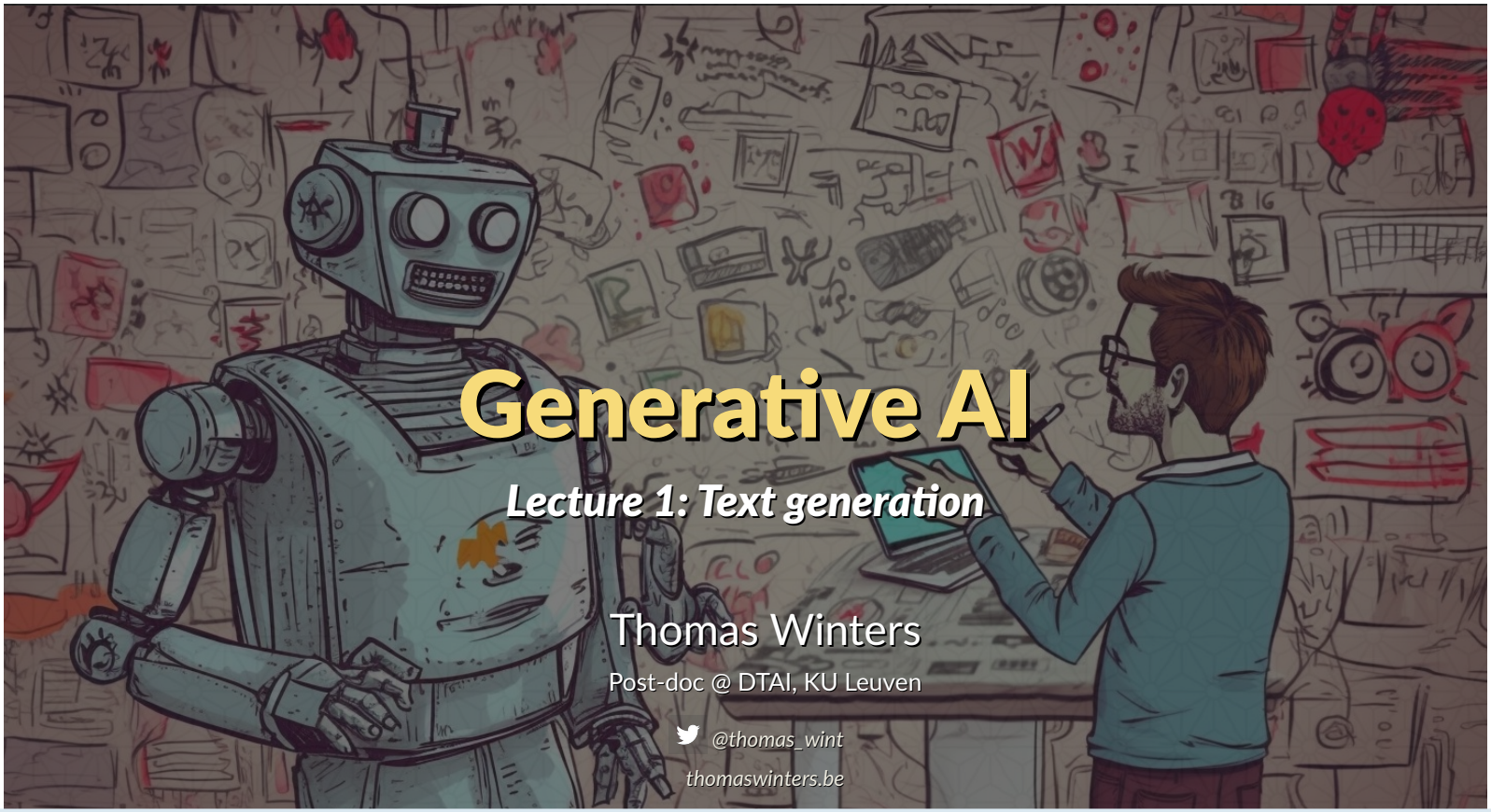
11. Question Answering



12. Q&A session

- Opportunity to ask any question about any of the lectures or the reading material.
- To be confirmed but possibly going through a mock exam.

13. Conversational agents



Not part of the course material but fun and informative lecture about LLM-based conversational agents.

Course overview

Linguistic structures: Why bother?

- POS tagging, dependency parsing, etc. are currently not used in production. But:
- LLMs are black boxes
  - Understanding the challenges in processing language is crucial to interpretability
  - It can help identifying weaknesses and analysing errors
- LLMs do not work well for low-resource languages
  - These tasks may still have relevance for these languages
- LLMs have massive amounts of parameters
  - More sustainable models may involve these tasks

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Views on teaching

Views

- Traditional lectures are bad - you learn best by activating your brain 🧠⛹️.
- Interaction with teacher and peers boosts learning - you should feel encouraged to come to class.
- You should not feel obliged to come (stay home if you are sick!)
- The belgian system where you do everything during the "blok" is bad, learning takes time

Course format

Course content

- All content material is online on the Toledo platform:
  - Pre-recorded videos of the lectures
  - Slides
  - Required readings
  - In-class exercises and solutions
  - Labs and solutions
- Slides (lectures/videos): high-level intuitive explanations
- Required readings: necessary to understand the models and algorithms in detail

Course format

In-person lectures

- 13 lectures: Thursdays 11h to 13h.
  - ⚠️ L6 Friday 30/10 9h30-11h30 in QDV-01.100
- Not recorded
- Same content as the videos
- Interactive, with short exercises
- Opportunity to ask for clarifications
- Get and give geedback
- Informal exam tips and preparation

Course format

Labs

- 7 exercises sessions of 2 hours, 2 groups.
- Mix of implementation and paper exercises (algorithms application).
- No additional material but direct preparation for the exam.
- Organized by the course's TAs

Relation to other courses this academic year

It is complementary to the *Computational Linguistics* course taught by Tim Van de Cruys (not a requirement).

Computational Linguistics

Stronger focus on the linguistic formalisms and high-level intuitions of models.

NLP

We go deeper in mathematical modelling and see the different algorithms in more detail.

We are collaborating to make the courses as compatible as possible. Redundancy may still happen. We welcome feedback!

Relation to other courses this academic year

Advanced NLP

taught by Pieter Delobelle.

- For MirAI students
- Large content overlap in the lectures + project
- In Dutch

Evaluation

Open-book written exam

Focus on understanding rather than memorizing.

Exam

- Theoretical part (50%):

  Broad overview questions, comparison of technologies, explaining algorithms.

- Exercise part (50%):

  Apply algorithms or evaluation metrics and analyse the results. Analyse the output of programs used in the labs.

Information sheet with example questions will be made available on toledo mid-October

Teaching team

Course coordinator and lecturer

Miryam de Lhoneux
Dpt of Computer Science
Celestijnlaan 200A, room 4.50
email: miryam.delhoneux@kuleuven.be

TAs

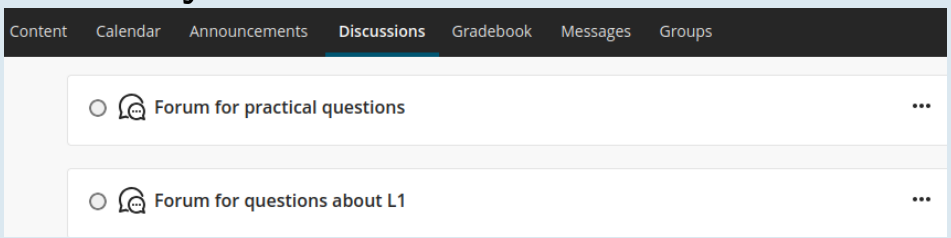
Coleman Haley coleman.haley@kuleuven.be
Wessel Poelman wessel.poelman@kuleuven.be
Benjamin Suter benjaminstefan.suter@ehu.eus

Please mind our time!

- There are ∼200 registered students
- Come or don't come to class, don't tell me about it
- Yes the required readings are required
- No you cannot bring your laptop/tablet to the exam

Questions

- You have a content-based question?
  - First try the forum on toledo



  - Or in the lectures or labs
- You have a practical question?
  - First check if the answer is on toledo
  - Then try the forum first
- Iff your question is urgent or goes unanswered for too long, then email
  - **Please include [H02B1A] in the subject line**
- Questions will be answered throughout the semester. No guarantee after that.

To sum up

- Let's try to decrease redundant communication
- And increase meaningful interaction

References

Ishan Jindal, Ranit Aharonov, Siddhartha Brahma, Huaiyu Zhu, and Yunyao Li. 2020. Improved semantic role labeling using parameterized neighborhood memory adaptation. *ArXiv*, abs/2011.14459.

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Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, Zhengxuan Wu, Bertie Vidgen, Grusha Prasad, Amanpreet Singh, Pratik Ringshia, Zhiyi Ma, Tristan Thrush, Sebastian Riedel, Zeerak Waseem, Pontus Stenetorp, Robin Jia, Mohit Bansal, Christopher Potts, and Adina Williams. 2021. Dynabench: Rethinking benchmarking in NLP. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.

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